

Today

- Linear Systems of Equations
- Gaussian Elim.
- Partial Pivoting

$$\begin{aligned} 2x_1 + x_2 &= 4 \\ -x_1 + x_2 &= 1 \end{aligned}$$

Unknowns: $x_1, \dots, x_n$

Linear Eqn.: $\sum_{j=1}^n a_{ij} x_j = b_i$

no nonlinear functions of x_j
 $x^2, e^x, \log(x), \sin(x)$ etc.

System of m
linear equations

$$\sum_{j=1}^n a_{ij} x_j = b_i \quad i = 1, \dots, m$$

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Book: $[A] \{x\} = \{b\}$

Handwriting: $A \vec{x} = \vec{b}$

for now $m = n$

HWO $\Rightarrow A \cdot x$

Lab

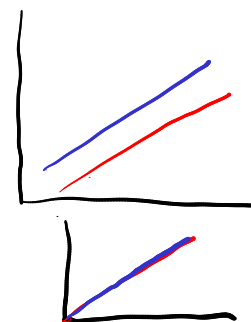
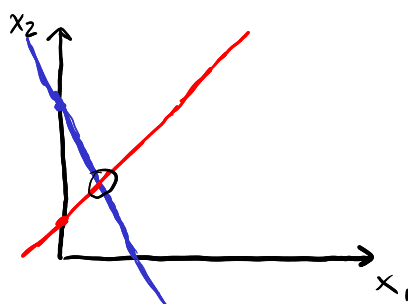
$Ax = b$
 Known $\uparrow$ Unknowns $\uparrow$ Known

$$\begin{aligned} 2x_1 + x_2 &= 4 \\ x_2 &= -2x_1 + 4 \end{aligned}$$

$$-x_1 + x_2 = 1$$

$$x_2 = x_1 + 1$$

$$\boxed{x_1 = 1, x_2 = 2}$$



Example:

$$\begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$3x_1 - x_2 = 0 \rightarrow 3x_1 - 3 = 0$$

$$2x_2 = 6$$

$$\boxed{x_2 = 3}$$

$$\boxed{x_1 = 1}$$

$$2x_1 + 3x_2 + 4x_3 = 19 \quad (1)$$

$$4x_1 + 11x_2 + 14x_3 = 55 \quad (2)$$

$$2x_1 + 8x_2 + 17x_3 = 50 \quad (3)$$

$$\begin{bmatrix} 2 & 3 & 4 & : & 19 \\ 4 & 11 & 14 & : & 55 \\ 2 & 8 & 17 & : & 50 \end{bmatrix}$$

$$\begin{aligned} & -2(1) \\ & -1(3) \end{aligned}$$

$$\begin{bmatrix} 2 & 3 & 4 & : & 19 \\ 0 & 5 & 6 & : & 17 \\ 0 & 5 & 13 & : & 31 \end{bmatrix}$$

$$-1(2)$$

$$\begin{bmatrix} 2x_1 + 3x_2 + 4x_3 = 19 \\ 0 \quad 5x_2 + 6x_3 = 17 \\ 0 \quad 0 \quad 7x_3 = 14 \end{bmatrix}$$

$$x_3 = 2$$

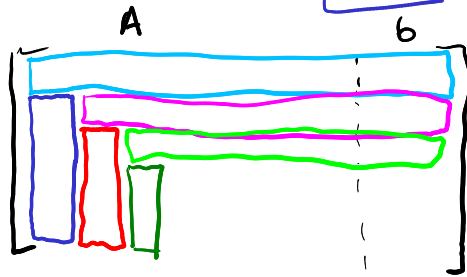
$$2x_1 + 3(1) + 4(2) = 19$$

$$2x_1 = 8$$

$$x_1 = 4$$

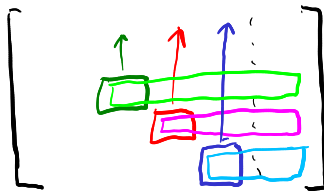
$$5x_2 + 12 = 17$$

$$x_2 = 1$$



Eliminate $\square$ with $\square$
Eliminate $\square$ with $\square$
etc

Back substitution



Solve for $\square$ with $\square$
Solve for $\square$ with $\square$
etc.

Example :

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 9 & -2 \\ 2 & 1 & -1 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 \\ 9 \\ 2 \end{bmatrix}$$

$$x = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

input : $n \times n$ matrix $[A]$, right-hand side $\{b\}$

output: solution $\{x\}$ of $[A]\{x\} = \{b\}$

// forward elimination

for $c \leftarrow 1$ to $n-1$ do

for $r \leftarrow c+1$ to n do

$f \leftarrow A[r, c] / A[c, c]$

$A[r, c:n] \leftarrow A[r, c:n] - f \cdot A[c, c:n]$

$b[r] \leftarrow b[r] - f \cdot b[c]$

end

end

// back substitution

for $r \leftarrow n, n-1, \dots, 1$ do

$s \leftarrow b[r]$

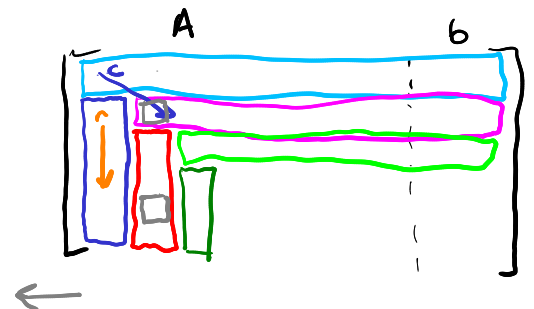
for $j \leftarrow r+1$ to n do

$s \leftarrow s - A[r, j] \cdot x[j]$

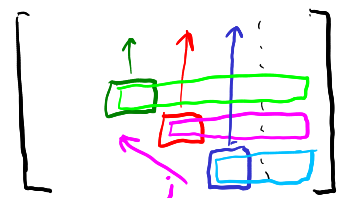
end

$x[r] \leftarrow s / A[r, r]$

end



Back substitution



Elimination
Backsub

$$A = \begin{bmatrix} 0 & 2 & 3 \\ 4 & 6 & 7 \\ 2 & -3 & 6 \end{bmatrix} \quad b = \begin{bmatrix} 8 \\ -3 \\ 5 \end{bmatrix}$$

$$f \leftarrow \underline{Aug[1,2]} / \underline{Aug[1,1]}$$

Divide by Zero

Solution: Partial Pivoting

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input :  $n \times n$  matrix  $[A]$ , right-hand side  $\{b\}$ 
output: solution  $\{x\}$  of  $[A]\{x\} = \{b\}$ 

// forward elimination
for  $c \leftarrow 1$  to  $n - 1$  do
    // find the largest entry in column  $c$ 
     $p \leftarrow \arg \max_{r \in c \dots n} |A[r, c]|$ 
    // swap that row into the pivot position
     $A[c, :] \leftrightarrow A[p, :]$ 
     $b[c] \leftrightarrow b[p]$ 
    for  $r \leftarrow c + 1$  to  $n$  do
         $f \leftarrow A[r, c] / A[c, c]$ 
         $A[r, c:n] \leftarrow A[r, c:n] - f \cdot A[c, c:n]$ 
         $b[r] \leftarrow b[r] - f \cdot b[c]$ 
    end
end

// back substitution
for  $r \leftarrow n, n - 1, \dots, 1$  do
     $s \leftarrow b[r]$ 
    for  $j \leftarrow r + 1$  to  $n$  do
         $s \leftarrow s - A[r, j] \cdot x[j]$ 
    end
     $x[r] \leftarrow s / A[r, r]$ 
end

```

